Aditya Shivale

Software Engineer

+91-8263033072 — shivaleaditya22@gmail.com — LinkedIn — GitHub

Summary

Backend-focused engineering undergraduate with experience building deployed full-stack applications, semantic retrieval systems, and REST-based backend architectures using Java, Node.js, MongoDB, and React.

Technical Skills

Languages: Java, JavaScript, SQL (MySQL)
Backend: Node.js, Express.js, REST APIs, Authentication, MVC Architecture
Frontend: React.js, EJS, HTML, CSS
Databases: MongoDB, MySQL
AI / Search Systems: Semantic Search, RAG, HNSW, Embeddings
Tools & Platforms: Git, Docker, Postman, Render, Cloudinary

Experience

Software / AI Intern **Oct 2025 – Nov 2025**
AICTE – Skills4Future (Shell India, Edunet Foundation) *Virtual*

- Engineered an end-to-end data processing pipeline with modular preprocessing and automated evaluation, reducing data transformation errors and ensuring reproducible training workflows.
- Designed modular components for structured dataset validation and experimentation workflows, improving maintainability and reducing engineering redundancy.
- Evaluated and optimized computer vision models using precision, recall, and mAP metrics; achieved a peak mAP50 score of 0.92 through iterative hyperparameter tuning and validation debugging.

Projects

Nexus – Semantic Vector Search & RAG System | *Java, React, Ollama, Docker* **2026**

Live Demo — GitHub

- Designed and deployed a multi-index semantic vector search engine supporting HNSW, KD-tree, and brute-force strategies for approximate nearest neighbor (ANN) retrieval.
- Implemented embedding-based retrieval and RAG pipelines using local Ollama models for semantic search and offline question answering.
- Developed interactive React dashboards for vector clustering visualization, retrieval benchmarking, and query latency analysis.

RoomScout – Full-Stack Accommodation Platform | *Node.js, Express.js, MongoDB, Passport.js* **2025**

Live Demo — GitHub

- Developed and deployed a full-stack accommodation platform featuring secure session authentication, ownership-based authorization, reviews, image uploads, and geolocation support.
- Engineered session-based authentication using Passport.js with MongoDB-backed persistent storage for protected route access and user session management.
- Designed MongoDB schemas with GeoJSON indexing, relational references, and Cloudinary-based media management.

Certifications

- **Oracle Cloud Infrastructure 2025** – Certified Generative AI Professional
- **AI & Data Analytics Internship Certificate** – AICTE

Education

B.E. – Artificial Intelligence & Data Science **2027**
DYPIEMR, Akurdi (Savitribai Phule Pune University) *CGPA: 8.4 / 10*